$$I = \int_{e^{-e}}^{e^{1/e}} \frac{(\lim_{n\to\infty} x \uparrow\uparrow n)^2}{x(e - \lim_{n\to\infty} x \uparrow\uparrow n)}\, dx, \quad \text{where} \quad x \uparrow\uparrow n \equiv \underbrace{x^{x^{x^{x^{x}}}}}_{n\text{-tower}}$$